Metals, Non-metals and Metalloids (p38-43)

Elements are the building blocks from which everything can be made. Elements are classified according to their properties as being metal, non-metal or metalloid.

Metals

1. Use the periodic table on page 38 to name ten metals.

i.

ii.

iii.

iv.

v.

vi.

vii.

viii.

ix.

x.

2. Metals share similar physical properties. Outline seven physical properties of metals.

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-
-
-
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3. Why are most metals not often used as pure elements?

4. In the table below, list the uses and properties of five pure metals.

Pure Metal	Symbol	Uses	Properties that make it suited to its use

5. What is an alloy?

6. What makes alloys useful?

7. Give an example of an alloy that has iron as its base metal. What makes it better than pure iron and what are some of its uses?

8. What element is added to iron to make steel?

9. What elements are added to steel to make stainless steel?

10. In the table below, list the composition, uses and advantages of four alloys.

Alloy	**Composition**	**Uses**	**Advantages**

Non-metals

1. Use the periodic table on page 38 to name 8 non-metals

i. v.

ii. vi.

iii. vii.

iv. viii.

2. Like metals, non-metals have similar physical properties to each other. Outline four physical properties of non-metals.

i.

ii.

iii.

iv.

3. Why is carbon an unusual element?

4. What percentage of known compounds contains carbon?

5. Describe the three forms, or allotropes, of carbon

i. Amorphous carbon:

ii. Graphite:

iii. Diamond:

Metalloids

1. Use the periodic table on page 38 to name 8 metalloids.

i.

ii.

iii.

iv.

v.

vi.

vii.

viii.

2. What is another name for the metalloids and why are they given this name?

3. Describe an important property of metalloids.

4. What are metalloids often used for?